

Abbreviation	Location	Study Period	Method	Statistical Approach	Exposure	Outcome Assessed	Effect Measure Modifier	Original Effect Measure Reported in Study	Climate
Aghababacian et al. (2019)	Dezful, Iran	March 2013 to March 2016	Time-Series	Generalized Linear Model (GLM) with Distributed Lag Model (DLM)	PM ₁₀	Cardiovascular and Respiratory Emergency Dispatches	Age & Sex	Relative Risk (RR) per IQR	BSh
Aghababacian et al. (2021)	Dezful, Iran	2014 to 2019	Time-Series	Distributed Lag Non-Linear Models (DLNM)	PM ₁₀	Non-accidental death (NAD, ICD-10: A00-R99) Cardiovascular death (CVD, ICD-10: I00-I99) Respiratory death (RD, ICD-10: J00-J99)	Age & Sex	Excess Risk (ER) per 10 µg/m3 exposure increase	BSh
Akbarzadeh et al. (2018)	Tehran, Iran	February 2014 to February 2016	Case-Crossover	Conditional logistic regression model	PM _{2.5} & PM ₁₀	Primary percutaneous coronary intervention (PCI)	Age & Sex	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk
Al-Taiar et al. (2014)	Kuwait (National)	January 1996 to December 2000	Time-Series	Generalized Additive Models (GAM)	PM ₁₀	Respiratory (ICD-9: 460-519) Cardiovascular diseases (ICD-9: 390–448) All-causes mortality (ICD ≤799)	Sex, Dust storm	Relative Risk (RR) per 10 µg/m3 exposure increase	BWh
Alahmad et al. (2023)	Kuwait (National)	January 1st, 2001 to December 31st, 2016	Time-Series	Generalized additive models (GAM) Two different lag structures: distributed lag models (DLM) and moving averages	PM _{2.5}	All-cause non-accidental (ICD-10: A00-R99) Cardiovascular (ICD-10: I00-I99) Respiratory (ICD-10: J00-J99)	Age & Sex	Relative Risk (RR) per 10 µg/m3 exposure increase	BWh
Albahar et al. (2022)	Kuwait (National)	1 January 2010 to 31 December 2018	Time-Series	Generalized linear models (GLMs) using Quasi-Poisson regression to account for overdispersion	PM _{2.5}	Respiratory (ICD-10: J00–J99) Asthma (ICD-10: J45)	Age & Sex	Percent Change (%) per 10 µg/m3 exposure increase	BWh

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AlBalawi et al. (2021)	Jubail, Saudi Arabia	January, 2007 to December, 2011	Time-Series	Generalized linear models (GLM) in Poisson regression	PM _{2.5} & PM ₁₀	Asthma hospital admissions	None	Relative Risk (RR) per 10 µg/m3 exposure increase	BWh
Bao et al. (2020)	Lanzhou, China	February 1, 2014 to December 31, 2017	Time-Series	Generalized Additive Models (GAM) in Poisson regression	PM _{2.5} & PM ₁₀	COPD (ICD-10: J40-J44)	Age & Sex	Percent Change (%) per 10 µg/m3 exposure increase	BSk
Basagaña et al. (2015)	Southern Europe	2003 to 2013	Case-Crossover	Conditional Poisson regression	PM _{2.5} & PM ₁₀	All non-external Mortality (ICD10: A00–R99) Cardiovascular Mortality (ICD-10: I00–I99) Respiratory Mortality (CD10: J00–J99)	None	Percent Change (%) per IQR increase	BSk
Bayart et al. (2024)	Ulaanbatar, Mongolia	January, 2016 to December, 2022	Case-Crossover	The conditional Poisson regression with distributed lag model (DLM)	PM _{2.5} & PM ₁₀	CVDs (ICD10: I00–I99), IHDs (ICD10: I20–I25) and cerebrovascular diseases, defined as total stroke (ICD10: I60–I69)	Age, Sex, & Season	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk
Borsi et al. (2020)	Ahvaz, Iran	March 20, 2008 to March 21, 2018	Time-Series	Generalized Linear Model (GLM) with Distributed Lag Model (DLM)	PM _{2.5} & PM ₁₀	Deep venous thrombosis (DVT)	Age & Sex	Relative Risk (RR) per 10 µg/m3 exposure increase	BSh
Bretón et al. (2020)	Monterrey, Mexico	1 January 2012 to 31 December 2015	Time-Series	Multiple Linear Regression (MLR) Principal Component Analysis (PCA) for PM source	PM ₁₀	Respiratory diseases (ICD-10: J00 to J99) Cardiovascular diseases (ICD-10: I00 to I99) All causes	Age & Sex	Relative Risk (RR) per 10 µg/m3 exposure increase	BSh

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Chai et al. (2019)	Lanzhou, China	1 January 2007 to 31 December 2016	Time-Series	Distributed Lag Non-Linear Models (DLNM)	PM _{2.5}	Respiratory diseases (ICD-10: J00-J99)	Age & Sex	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk
Chen et al. (2017)	Tianjin, China Urumqi, China Taiyuan, China Hohhot, China Xining, China Baotou, China	October 28, 2013 to December 31, 2014	Case-Crossover	quasi-Poisson regression with a constrained distributed lag model (CDLM)	PM _{2.5} & PM ₁₀	Emergency hospital visits	Age & Sex	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk
Chen et al. (2020) a	Lanzhou, China	January 1, 2017 to December 31, 2018	Time-Series	Generalized Additive Models (GAM)	PM _{2.5} & PM ₁₀	Cardiovascular disease (ICD-10: I00-I99)	Age, Sex, & Season	Percent Change (%) per 10 µg/m3 exposure increase	BSk
Chen et al. (2020) b	Tehran, Iran	January 1, 2012 to December 31, 2016	Time-Series	Standard time-series generalized model (GLM)	PM _{2.5}	Cardiovascular disease (ICD-10: I00-I99) Ischemic heart disease (IHD, ICD-10: I20-I25) Stroke (ICD-10: I60-I69) Sudden cardiac death (SCD, ICD-10: I46)	Age & Sex	Percent Change (%) per IQR increase	BSk
Chen et al. (2020) c	Hohhot, China Urumqi, China	October 28, 2013 to December 31, 2014	Case-Crossover	Constrained Distributed Lag Model	PM ₁ , PM _{2.5} & PM ₁₀	Ischemic stroke (I63)	Age, Sex, & Season	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk
Chen et al. (2021)	Kuwait (National)	January 1, 2000, to December 31, 2016,	Time-Series	quasi-Poisson regression	PM _{2.5}	All-cause, Cardiovascular, Respiratory Mortality	Wildfire	Rate Ratios (RR) per 10 µg/m3 exposure increase	BWh

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Cheng et al. (2021)	Lanzhou, China	January 1, 2013 to July 31, 2017	Time-Series	Generalized Additive Models (GAM)	PM _{2.5} & PM ₁₀	Respiratory disease	Age, Sex, & Season	Relative Risk (RR) per IQR increase	BSk
Cirera et al. (2012)	Cartagena, Spain	January, 1995 to December, 1998	Time-Series	Poisson regression with Generalised Additive Models	TSP	Asthma, COPD	None	Relative Risk (RR) per 10 µg/m3 exposure increase	BSh
Colonna et al. (2023)	Kuwait (National)	October 25th, 2017 to November 3rd, 2019	Case-Crossover	Generalized linear model (GLMs)	PM _{2.5}	Respiratory disease (ICD-10: J00-J99)	Age & Sex	Percent Change (%) per IQR increase	BWh
Cordova et al. (2020)	Lima, Peru	2011 to 2015	Time-Series	Negative binomial model (the Poisson model was over-dispersed)	PM _{2.5}	Acute lower respiratory disease (ALRI, ICD-10: J00-J11.1), Pneumonia (ICD-10: J12.0-J18.9), Acute bronchiolitis/asthma (ICD-10: J21.0-J21.9 / J44.8–46.0)	Age	Rate Ratios (RR) per IQR increase	BWh
Dales et al. (2020)	Santiago, Chile	January 2001 to December 2012	Time-Series	Generalised linear model (GLM)	PM _{2.5}	Idiopathic pulmonary fibrosis (IPF, ICD-10: J84.1)	Age & Sex	Relative Risk (RR) per IQR increase	BSk
Dastoorpoor et al. (2019)	Ahvaz, Iran	January, 2008 to December, 2018	Time-Series	Generalized Linear Model (GLM) with Distributed Lag Model (DLM)	PM ₁₀	Acute rheumatic fever (I00–I02), Chronic rheumatic heart diseases (I05–I09), Hypertensive diseases (I10–I16), Ischemic heart diseases (I20–I25), Pulmonary heart disease and diseases of pulmonary circulation (I26–I28), Other forms of heart disease (I30–I52), Cerebrovascular diseases (I60–I69), Diseases of arteries, arterioles and capillaries (I70–I79), Diseases of veins, lymphatic vessels and lymph nodes (I80–I89), not elsewhere classified, Other and unspecified disorders of the circulatory system (I95–I99)	Age & Sex	Risk Ratio (RR) for 10 µg/m3 increase of exposure	BSh

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de Bont et al. (2024)	Delhi, India	January, 2011 to December, 2018	Time-Series	Generalized Additive Models (GAM) in Poisson regression	PM _{2.5}	All Cause Mortality	None	Percent Change (%) per 10 µg/m3 exposure increase	BSh
Diaz et al. (2012)	Madrid, Spain	January, 2003 to December, 2005	Case-Crossover	Poisson Regression Model	PM ₁₀	Organic causes except accidents (ICD-10: A00-R99), respiratory causes (ICD-10: J00-J99), circulatory causes (ICD-10: I00-I99) and cerebrovascular mortality (ICD-10: I60-I69)	Season & Dust storm	IR%, for 10 µg/m3 increase of exposure	BSk
Do et al. (2024)	California, USA	2006 to 2019	Case-Crossover	Weighted conditional logistic regression & Spatial Bayesian Hierarchical Model	PM _{2.5}	Respiratory acute care utilizations	None	Odds Ratio (OR)	BSk
Dominguez-Rodriguez et al. (2021)	Tenerife, Spain	December, 2012 to December, 2017	Case-Crossover	Conditional logistic regression model	PM _{2.5} & PM ₁₀	acute coronary syndrome	None	Incidence Rate Ratios (IRR) for 10 µg/m3 increase of exposure	BSh
Dong et al. (2023)	Lanzhou, China	January 1, 2013 to December 31, 2019	Time-Series	Overdispersed generalized additive model (quasi-likelihood) with natural smooth functions	PM _{2.5} & PM ₁₀	AECOPD (ICD-10: J44.0-J44.9)	Age, Sex, & Season	Percent Change (%) per 10 µg/m3 exposure increase	BSk
Dutta & Jinsart (2022)	Delhi, India	1 January 2016 to 31 December 2018	Time-Series	Generalized Additive Models (GAM)	PM _{2.5} & PM ₁₀	Respiratory diseases (ICD-10: J00-J99)	Season	Relative Risk (RR) per 10 µg/m3 exposure increase	BSh

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Faustini et al. (2015)	Madrid, Spain	2003 and 2010	Case-crossover	Poisson Regression model	PM ₁₀	Cardiovascular Mortality (ICD-10: I00-I99) Respiratory Mortality (ICD-10: J00-J99)	None	Percent Change (%) per 10 µg/m3 exposure increase	BSk
Gouveia et al. (2018)	Santiago, Chile	1997 to 2005	Time-Series	Generalized Additive Models (GAM) in Poisson regression	PM ₁₀	Respiratory disease mortality (ICD-10: J00-J98), Lower respiratory infections (LRI, ICD-10: J10-J22),	Age & Season	Percent Change (%) per 10 µg/m3 exposure increase	BSk
Horne et al. (2018)	Wasatch Front, USA	June, 1905 to July, 1905	Case-Crossover	Conditional logistic regression model	PM _{2.5}	ALRI or the primary diagnosis of respiratory failure [ICD-10] J96.00)	Age	Odds Ratio (OR) per 10 µg/m3 exposure increase	BSk
Horne et al. (2024)	Wasatch Front, USA	January, 1999 to March, 2022	Case-Crossover	Conditional Poisson regression using a quasi-Poisson distribution	PM _{2.5}	Asthma (ICD-9: 493; ICD-10: J45) or COPD (ICD-9: 491, 492, 494, and 496; ICD-10: J41, J42, J43, J44, and J47)	Wildfire	Odds Ratio (OR) per 10 µg/m3 exposure increase	BSk
Hutchinson et al. (2018)	San Diego, USA	August 1 2007 to December 31, 2007	Case-Crossover	conditional logistic regressions (SAS procedure LOGISTIC)	PM _{2.5}	All Respiratory and Cardiovascular (ICD-9)	Age, Wildfire	Rate Ratios (RR) per 10 µg/m3 exposure increase	BSk
Ji et al. (2023)	Lanzhou, China	January 1, 2014 to December 31, 2019	Time-Series	Generalized additive model & Distributed lag nonlinear model (DLNM)	PM _{2.5} & PM ₁₀	Allergic rhinitis (ICD-10: J30)	Age, Sex, & Season	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk

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Jin et al. (2023)	Lanzhou, China	January 1st, 2014 to December 31st, 2019	Time-Series	Distributed Lag Non-Linear Models (DLNM)	PM _{2.5} & PM ₁₀	Respiratory diseases (ICD-10: J00-J99)	Age & Sex	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk
Joshi et al. (2022)	Delhi, India	1 January 2013 to 31 December 2016	Time-Series	Semiparametric quasi-Poisson time series regression model	PM _{2.5}	All cause Mortality	Age, Sex, & Season	Percent Change (%) per IQR increase	BSh
Kim et al. (2012)	Denver, USA	January, 2003 to December, 2007	Time-Series	Generalised linear model (GLM)	PM _{2.5}	Cardiovascular hospital admissions (codes 390–459) and respiratory hospital admissions (codes 460–519)	None	Relative Risk (RR) per IQR increase	BSk
Kim et al. (2013)	Denver, USA	January, 2003 to December, 2007	Time-Series	Generalised linear model (GLM)	PM _{2.5}	Cardiovascular, respiratory disease, ischemic heart disease (410–414), congestive heart failure (428), cerebrovascular disease (430–438), ischemic stroke (434.01, 434.11, 434.91, 436), chronic obstructive pulmonary disease (COPD: 490–492, 496), asthma (493), pneumonia (480–486), and upper respiratory infection (460–466, 477)	None	Relative Risk (RR) per IQR increase	BSk
Kim et al. (2017)	Urumqi, China Tianjin, China Lanzhou, China	1993 to 2009	Time-Series	Two-stage Bayesian hierarchical statistical model	PM ₁₀	All-natural-cause mortality	Season	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk
Kiser et al. (2020)	Reno, USA	January, 2013 to December, 2018	Time-Series	Generalized Additive Models (GAM)	PM _{2.5} & PM ₁₀	Asthma ICD-10 codes J45.20-J45.998	Wildfire	Rate Ratios (RR) per 5 µg/m3 exposure increase	BSk

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Leibel et al. (2020)	San Diego, USA	2011 to 2017	Time-Series	Poisson Regression Model	PM _{2.5}	Respiratory visits	Age, Wildfire	Excess Daily Visits (% Increase)	BSk
Leogrande et al. (2019)	Taranto, Italy	2008 to 2014	cohort	multiple Poisson regression analysis	PM ₁₀	Natural causes (ICD-9: 001–799, ICD-10: A00–R99), Circulatory system (ICD-9: 390–459, ICD-10: I00–I99), Heart diseases (ICD-9: 390–429, ICD-10: I00–I51), Respiratory diseases (ICD-9: 460–519, ICD-10: J00–J99)	Age	Percent Change (%) per 1 µg/m3 exposure increase	BSk
Li et al. (2016)	Tianjin, China	January 2002 to December 2006	Time-Series	Generalised additive model (GAM)	PM ₁₀	Ischaemic heart disease mortality (ICD-10: I20–I25)	Age & Sex	Percent Change (%) per IQR increase	BSk
Li et al. (2021)	Xining, China	January 1, 2013, to December 31, 2018	Time-Series	Generalized Additive Model (GAM), with a quasi-Poisson link	PM _{2.5} & PM ₁₀	All respiratory diseases (ICD-10: J00–J99), Acute upper respiratory infections (ICD-10: J00–J06) and Acute lower respiratory infections (ICD-10: J10–J18, J20–J22 and J40)	Age & Sex	Percent Change (%) per 10 µg/m3 exposure increase	BSk
Liu et al. (2019)	Shijiazhuang, China	January, 2013 to December, 2016	Time-Series	Generalized additive models combined with penalized distributed lag nonlinear models	PM _{2.5} & PM ₁₀	Asthma (International Classification of Diseases, ICD-9: J45.000, J45.003, J45.007, J45.100, J45.800, J45.900, J45.902, J45.903, J45.904, J46.x 00)	Season	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk
Liu et al. (2020)	Lanzhou, China	2004–2009 and 2014–2017 to	Time-Series	Generalized Additive Models (GAM) in Poisson regression	PM ₁₀	Non-accidental causes (ICD-10: A00–R99) Circulatory diseases (ICD-10: I00–I99) Respiratory diseases (ICD10: J00–J99)	Age & Sex	Percent Change (%) per 10 µg/m3 exposure increase	BSk

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Liu et al. (2022) a	Lanzhou, China	1 February 2014 to 31 December 2018	Time-Series	Generalized Additive Models (GAM)	PM _{2.5} & PM ₁₀	Upper respiratory tract infection (URTI, ICD-10: J00-J06 and J30-J39)	Age, Sex, & Season	Percent Change (%) per 10 µg/m3 exposure increase	BSk
Liu et al. (2022) b	Lanzhou, China	2013 to 2017	Time-Series	Distributed Lag Non-Linear Models (DLNM)	PM _{2.5} & PM ₁₀	hypertension	Age, Sex, & Season	Rate Ratios (RR) per 10 µg/m3 exposure increase	BSk
Liu et al. (2023)	Lanzhou, China	1 Jan 2013 to 31 Dec 2020	Time-Series	Distributed Lag Non-Linear Models (DLNM)	PM _{2.5} & PM ₁₀	Coronary Heart Disease (ICD-10: I20–I25)	Age, Sex, & Season	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk
Lopez-Villarrubia et al. (2012) a	Las Palmas de Gran Canaria, Spain	January, 2001 to December, 2004	Time-Series	Generalized Additive Models (GAM)	PM _{2.5} & PM ₁₀	daily mortality from all heart diseases (ICD-10: I20-I25, I46-I50, R98-99, R09.2) and daily mortality from all respiratory diseases (ICD-10: J00-J95, J97-J99, R09.1)	None	Percent Change (%)	BWh
Lopez-Villarrubia et al. (2012) b	Las Palmas de Gran Canaria, Spain	January, 2001 to December, 2004	Time-Series	Generalized Additive Models (GAM)	PM _{2.5} & PM ₁₀	ARSD [ICD-9: 460–519], COPD [ICD-9: 490–492, 494–496] and asthma [ICD-9: 493]	Dust storm	Percent Change (%)	BWh
Lu et al. (2023)	Baotou, China	2015 to 2019	Time-Series	Generalized Additive Models (GAM)	PM _{2.5} & PM ₁₀	Non-accidental deaths (ICD-10: A00-R99) Cardiovascular deaths (ICD-10: I00-I99) Respiratory deaths (ICD-10: J00-J99)	Age & Sex	Excess Risk (ER) per 10 µg/m3 exposure increase	BSk

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Ma et al. (2016)	Lanzhou, China	1 March 2007 to 31 May 2011	Time-Series	Poisson generalized additive model (GAM)	PM ₁₀	Total diseases (ICD-10: A00-R99) Respiratory diseases (ICD-10: J00-J99)	Age & Sex	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk
Ma et al. (2020)	Lanzhou, China	January 1st, 2014 to December 31st, 2016	Time-Series	Quasi-Poisson regression with generalized additive model (GAM)	PM _{2.5} & PM ₁₀	Respiratory diseases (ICD-10: J00-J99) Subcategories Pneumonia (ICD-10: J12-J18); Asthma (ICD-10: J45-J46); Bronchitis and upper respiratory tract infection (ICD-10: J00-J06, J20-J21, J30-J39, J40-J42); and other respiratory diseases (ICD-10: J22, J43-J44, J47, J60-J99)	Age	Percent Change (%) per 10 µg/m3 exposure increase	BSk
Madaniyazi et al. (2024)	Lima, Peru	January, 2010 to December, 2014	Time-Series	generalized linear models with quasi-Poisson family	PM _{2.5} & PM ₁₀	Non-external causes (International Classification of Diseases, 9th revision (ICD-9) codes 0 to 799 and ICD-10 codes A0 to R99	None	Relative Risk (RR) per 10 µg/m3 exposure increase	BWh
Malamardi et al. (2023)	Mysore, India	1 January 2017 to 31 December 2021	Time-Series	Generalised linear regression models (GLMs) if GAM not sufficient Generalised Additive Models (GAMs)	PM _{2.5} & PM ₁₀	Asthma and WALRI hospital admissions (ICD-10: J45, J46, or R06.2)	Age & Sex	Relative Risk (RR) per 10 µg/m3 exposure increase	BSh
Mohammadian-Khoshnoud et al. (2023)	Hamadan, Iran	March 2016 to September 2020	Time-Series	Zero-inflated Poisson Model	PM _{2.5} & PM ₁₀	Acute myocardial infarction	Sex	Beta (from Zero-inflated Poisson model)	BSk
Motesaddi et al. (2022)	Tehran, Iran	21 April 2014 to 22 August 2017	Time-Series	Distributed Lag Non-Linear Models (DLNM)	PM _{2.5}	Total cardiovascular diseases (ICD-10: I00–I99) Ischemic heart diseases (IHD, ICD-10: I20–I25) Cerebrovascular diseases (ICD-10: I60–I69) Other cardiovascular diseases (ICD-10: I00–I16, I26–I52, I70–I99)	Age & Sex	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk

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Nayebare et al. (2019)	Jeddah, Saudi Arabia	Apr 8th, 2013 to Feb 18th, 2014	Time-Series	Generalized linear model (GLMs)	PM _{2.5}	Cardiovascular (ICD-10: I00-I99) Respiratory diseases (ICD-10: J00-J99)	Age & Sex	Relative Risk (RR) per 10 µg/m3 exposure increase	BWh
Neophytou et al. (2013)	Nicosia, Cyprus	January, 2004 to January, 2008	Time-Series	Generalized Additive Models (GAM)	PM ₁₀	Cardiovascular and respiratory daily mortality	Dust storm	Percent Change (%) per 10 µg/m3 exposure increase	BSh
Ni et al. (2022)	Tianjin, China	January 1, 2015, to December 31, 2019	Time-Series	generalized additive model (GAM) with a Gaussian link	PM _{2.5} & PM ₁₀	Total respiratory diseases (ICD-10: RD: J00-J99) Chronic obstructive pulmonary diseases (ICD-10: COPD: J41-J44)	Sex	The daily avoidable YLL per one million population	BSk
Ning et al. (2024) a	Xining, China	January, 2016 to December, 2021	Case-Crossover	Conditional logistic regression model	PM _{2.5} & PM ₁₀	Respiratory disease (ICD-10: J00-J99)	Age & Sex	Odds Ratio (OR) per 10 µg/m3 exposure increase	BSk
Ning et al. (2024) b	Xining, China	January, 2016 to December, 2021	Case-Crossover	Distributed Lag Non-Linear Models (DLNM)	PM _{2.5}	Respiratory disease (ICD-10: J00-J99)	Age & Sex	Percent Change (%) per 10 µg/m3 exposure increase	BSk
Ning et al. (2024) c	Xining, China	January, 2016 to December, 2021	Case-Crossover	Conditional Poisson regression model	PM ₁₀	Ischemic heart disease (IHD)	Age & Sex	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk

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Ostro et al. (2016)	El Cajon, USA	2005 to 2009	Case-Crossover	Conditional logistic regression model	PM _{2.5}	Cardiovascular Disease (ICD-10: I00 to I99)	Age	Excess Risk (ER) per 10 µg/m3 exposure increase	BSh
Pirozzi et al. (2018)	Wasatch Front, USA	December, 2009 to November, 2012	Case-Crossover	Conditional logistic regression model	PM _{2.5}	Pneumonia	Age & Season	Odds Ratio (OR) per 10 µg/m3 exposure increase	BSk
Pope et al. (2015)	Wasatch Front, USA	October, 1993 to May, 2014	Case-Crossover	Conditional logistic regression model	PM _{2.5}	Acute coronary syndrome events	None	Odds Ratio (OR) per 10 µg/m3 exposure increase	BSk
Qi et al. (2020)	Tianjin, China	1 April 2018 to 31 March 2019	Case-Crossover	Spearman's correlation tests and conditional logistic regression	PM _{2.5} & PM ₁₀	Ischemic stroke (ICD-10: I67.81)	Age	Odds Ratio (OR) per IQR increase	BSk
Resnick et al. (2015)	Albuquerque, USA	May 1, 2011 to July 8, 2011	Does not say	Poisson Regression Model	PM _{2.5}	Specific respiratory or cardiovascular diagnosis in the first 3 diagnostic fields of their patient ED record	Age	Relative Rate (RR)	BSk
Romieu et al. (2012)	Santiago, Chile & Monterrey, Mexico	1997 to 2005	Time-Series	Generalized linear models (GLM) in Poisson regression	PM ₁₀	All-natural-cause (ICD-9: 001–799) (ICD-10: A00–T98, Z00–Z99), Cardiovascular (ICD-9: 390–459) (ICD-10: I00–I99), Cardiopulmonary (ICD-9: 390–459, 460–519), (ICD-10: I00–I99, J00–J98), Cerebrovascular–stroke (ICD-9: 430–438), (ICD-10: I60–I69), Respiratory (ICD-9: 460–519), (ICD-10: J00–J98), COPD (ICD-9: 490–492, 494–496), (ICD-10: J40–J44, J46–J47), Lower respiratory infection (ICD-9: 466, 480, 487), (ICD-10: J10, J22)	Age	Relative Risk (RR) per 10 µg/m3 exposure increase	BSh

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Rosenquist et al. (2020)	Reno–Sparks, USA	January 2011 to June 2017	Case-Crossover	single-pollutant distributed lag nonlinear models (DLNM)	PM _{2.5}	Asthma (ICD-10: J45.*)	Age, Sex, & Season	Odds Ratio (OR) per 10 µg/m3 exposure increase	BSk
Samoli et al. (2013)	Madrid, Spain	January, 2007 to December, 2009	Time-Series	Poisson Regression Model, allowing for overdispersion	PM _{2.5} & PM ₁₀	all-cause mortality, cardiovascular mortality (ICD-9 codes 390–459, ICD10 codes I00–I99) and respiratory mortality (ICD-9 codes 460–519, ICD10 codes J00–J99)	Age	Percent Change (%) per 10 µg/m3 exposure increase	BSk
Shahsavani et al. (2020)	Ahvaz, Iran	March, 2015 to September, 2017	Case-Crossover	Conditional Poisson regression model	PM _{2.5} & PM ₁₀	all-natural cause mortality (ICD 10th revision: A00–R99)	Dust storm	Percent Change (%) per 10 µg/m3 exposure increase	BSh
Shen et al. (2023)	Lanzhou, China	January 1, 2013 to December 31, 2016	Time-Series	Generalized Additive Models (GAM) in Poisson regression	PM _{2.5} & PM ₁₀	Respiratory diseases (ICD-10: J00–J99)	Age, Sex, & Season	Relative Risk (RR) per IQR increase	BSk
Song et al. (2018)	Shijiazhuang, China	January 1, 2013 to December 31, 2015	Time-Series	Generalized Additive Models (GAM)	PM _{2.5} & PM ₁₀	Respiratory disease (ICD-10: J00–J99)	Age & Season	Percent Change (%) per 10 µg/m3 exposure increase	BSk
Song et al. (2019)	Shijiazhuang, China	January 1, 2013 and December 31, 2016	Time-Series	over-dispersed generalized additive model (GAM)	PM _{2.5} & PM ₁₀	non-accident hospital admissions of patients with hypertension	Age, Sex, & Season	Percent Change (%) per 10 µg/m3 exposure increase	BSk

Abbreviation	Location	Study Period	Method	Statistical Approach	Exposure	Outcome Assessed	Effect Measure Modifier	Original Effect Measure Reported in Study	Climate
Steenland et al. (2022)	Lima, Peru	March, 2010 to December, 2016	Time-Series	Generalized linear models (GLMs) with quasi-Poisson regression (correcting for overdispersion)	PM _{2.5}	All-cause, cardiovascular (ICD codes I00-I99), and respiratory (ICD codes J00-J99) mortality, and cardiovascular and respiratory causes for ER visits	None	Rate Ratios (RR) per 10 µg/m3 exposure increase	BWh
Sun et al. (2024)	Tianjin, China	1 January 2019, to 31 December 2020	Case-Crossover	Distributed Lag Non-linear Model (DLNM), Conditional logistic regression	PM _{2.5} & PM ₁₀	Ischemic Cerebrovascular Event (ICD-10: I63, G45)	Age & Sex	Odds Ratio (OR) per IQR increase	BSk
Tabaghi et al. (2023)	Tehran, Iran	July 2017 to March 2019	Case-Crossover	Conditional logistic regression model	PM _{2.5} & PM ₁₀	Acute myocardial infarction	Age & Sex	Odds Ratio (OR) per 10 µg/m3 exposure increase	BSk
Tapia et al. (2020)	Lima, Peru	March 1, 2010 to December 30, 2016	Time-Series	Generalized Linear Models (GLMs) in Poisson regression	PM _{2.5}	Respiratory diseases (RD, ICD-10: J00–J45) Ischemic heart disease (ICD-10: I20–I25) Stroke (G45, I63–I67)	Age	Rate Ratios (RR) per IQR increase	BWh
Tong et al. (2014)	Tianjin, China	1 January 2008 to 31 November 2011	Time-Series	Generalized linear model with natural splines	PM ₁₀	Cardiovascular morbidity (ICD-10: cerebral infarction (63.9), primary diagnosed hypertension (I10), cerebral hemorrhage (S06.369A), acute myocardial infarction (I21.9), and subarachnoid hemorrhage (I60.9))	Age & Sex	Percent Change (%) per 10 µg/m3 exposure increase	BSk
Valdés et al. (2012)	Santiago, Chile	1998 to 2007	Time-Series	Poisson regression in a generalized additive model	PM _{2.5}	Respiratory diseases (ICD-9: 460-519) Cardiovascular diseases (CVD, ICD-9: 390-429) Chronic obstructive pulmonary diseases (COPD, ICD-9: 490-496) Cerebrovascular diseases (cerebro, ICD-9: 430-459)	Age	Percent Change (%) per 10 µg/m3 exposure increase	BSk

Abbreviation	Location	Study Period	Method	Statistical Approach	Exposure	Outcome Assessed	Effect Measure Modifier	Original Effect Measure Reported in Study	Climate
Vodonos et al. (2015)	Be'er-Sheva, Israel	2001 to 2010	Case-Crossover	Generalized additive Poisson regression models (GAM) with Conditional logistic regression model	PM ₁₀	Acute coronary syndrome (ACS, ICD-9: 411.81) Unstable angina (UA, ICD-9: 411.1) Acute myocardial infarction (AMI, ICD-9: 410)	Age & Sex	Odds Ratio (OR) per 10 µg/m3 exposure increase	BSh
Vu et al. (2021)	Lima, Peru	March 1, 2010 to December 30, 2016	Time-Series	Generalized Linear Models (GLMs) in Poisson regression	PM _{2.5}	Asthma (ICD-10: J45-J46)	Age & Sex	Rate Ratios (RR) per IQR increase	BWh
Wang et al. (2013)	Tianjin, China	January, 2006 to December, 2010	Time-Series	Generalized Additive Models (GAM)	PM ₁₀	non-accidental deaths (A00–R99; International Classification of Diseases, 10th Revision), cardiovascular deaths (I00–I99), and cardiopulmonary deaths (I00–I99, J00–J99)	None	Percent Change (%) per 10 µg/m3 exposure increase	BSk
Wang et al. (2023)	Hohhot, China	January, 2018 to December, 2020	Time-Series	Generalized Additive Models (GAM), Spearman Correlation Analysis	PM _{2.5} & PM ₁₀	Allergic rhinitis ICD-10 (J30.401)	None	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk
Wu et al. (2019)	Lanzhou, China	January, 2014 to December, 2017	Time-Series	Quasi-Poisson regression model combined with a distributed lag non-linear model (DLNM)	PM _{2.5} & PM ₁₀	Cardiovascular diseases (ICD-10 codes I00–I99) and cause-specific diseases, including cerebrovascular disease (CD, ICD-10 codes I60–I69), ischemic heart disease (IHD, ICD-10 codes I20–I25), heart rhythm disturbances (HRD, ICD-10 codes I44–I49), and heart failure (HF, ICD-10 code I50)	None	Excess Risk (ER) per 10 µg/m3 exposure increase	BSk
Yadav et al. (2023)	Delhi, India	June, 2017 to February, 2019	Time-Series	Generalised additive model (GAM) with the Poisson link function	PM _{2.5}	Acute respiratory symptoms	None	Percent Change (%) per 10 µg/m3 exposure increase	BSh

Abbreviation	Location	Study Period	Method	Statistical Aproach	Exposure	Outcome Assessed	Effect Measure Modifier	Original Effect Measure Reported in Study	Climate
Yin et al. (2017)	Tianjin, China Yinchuan, China Urumqi, China Taiyuan, China Hohhot, China Xining, China	1 January 2010 to 29 June 2013	Time-Series	Generalised linear model (GLM)	PM ₁₀	Cardiovascular (ICD-10: I00-99) Respiratory (ICD-10: J00-99) All other diseases	Age & Sex	Percent Change (%) per 10 µg/m3 exposure increase	BWk
Yin et al. (2022)	Tehran, Iran	January 1, 2012 to December 31, 2016	Time-Series	Quasi-Poisson or Gaussian time-series regression	PM _{2.5}	Non-accidental mortality (ICD-10: A00-R99)	Age & Sex	Years Life Lost (YLL)	BSh
Yitshak-Sade et al. (2015)	Be'er-Sheva, Israel	2005 to 2012	Case-Crossover	Conditional logistic regression model	PM _{2.5} & PM ₁₀	Hemorrhagic stroke (ICD-9: 430–432) Ischemic stroke (ICD-9: 432–435)	Sex	Odds Ratio (OR) per IQR increase	BSh
Yitshak-Sade et al. (2017)	Be'er-Sheva, Israel	2003 to 2013	Case-Crossover	Conditional logistic regression model	PM _{2.5} & PM ₁₀	Bronchiolitis (ICD-9: 466)	Sex	Odds Ratio (OR) per IQR increase	BSh
You et al. (2018)	Imperial County, USA	2004 to 2007	Case-Crossover	Cox Proportional Hazard model	PM _{2.5}	Non-accidental mortality (ICD-10: A00-R99)	Age	Odds Ratio (OR)	BWh
Yu et al. (2024)	Lanzhou, China	1 January 2013 to 31 December 2020	Time-Series	Generalized additive model (GAM) with a quasi-Poisson distribution to construct DLNM	PM _{2.5} & PM ₁₀	Cardiovascular Disease (ICD-10: I00 to I99)	Age, Sex, & Season	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk

Abbreviation	Location	Study Period	Method	Statistical Approach	Exposure	Outcome Assessed	Effect Measure Modifier	Original Effect Measure Reported in Study	Climate
Zeng et al. (2015)	Tianjin, China	January 1, 2007, to December 31, 2009	Time-Series	Generalized Additive Models (GAM) in Poisson regression	PM ₁₀	nonaccidental (ICD-10: A00-R99), cardiovascular (ICD-10: I00-I99), or respiratory system (ICD-10: J00-J99)	None	Exposure-Response Relationship	BSk
Zeng et al. (2017)	Tianjin, China	January 1st, 2001 to December 31st, 2010	Time-Series	Generalized Additive Models (GAM)	PM ₁₀	Non-accidental death (ICD-10: A00-R99)	Age	Excess Risk (ER) per 10 µg/m3 exposure increase	BSk
Zhai et al. (2021)	Lanzhou, China	1 February 2014 to 31 December 2017	Time-Series	Semi-parametric generalized additive model (SGAM)	PM _{2.5} & PM ₁₀	Respiratory diseases (ICD-10: J00-J99). Upper respiratory tract infection (URTI; ICD-10: J00-J06, J30-J39) Lower respiratory tract infection (LRTI; ICD-10: J20-J22 and J40-J47) Asthma (ICD-10: J45-J46)	Age & Sex	Excess Risk (ER) per 10 µg/m3 exposure increase	BSk
Zhang et al. (2023) a	Lanzhou, China	January 1, 2014 to December 31, 2020	Time-Series	Generalized linear models and distributional lag models	PM _{2.5}	Acute lower respiratory infections (ICD-10: J12-J18 and J20-J22) Pneumonia (ICD-10: J12-J18) Bronchiolitis (ICD-10: J20-J21)	Age, Sex, & Season	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk
Zhang et al. (2023) b	Lanzhou, China	January 1, 2015 to December 31, 2020	Time-Series	Generalized additive model & Distributed lag nonlinear model (DLNM)	PM _{2.5}	Heart Failure (ICD-10: I50)	Age, Sex, & Season	Relative Risk (RR) per 10 µg/m3 exposure increase	BSk
Zhao et al. (2022)	Urumqi, China	2013 to 2016	Time-Series	generalized additive model (GAM) with a Poisson link	PM _{2.5}	all non-accidental causes (ICD-10, A00-R99)	Season	Percent Change (%) per 10 µg/m3 exposure increase	BSk

Abbreviation	Location	Study Period	Method	Statistical Approach	Exposure	Outcome Assessed	Effect Measure Modifier	Original Effect Measure Reported in Study	Climate
Zheng et al. (2013)	Lanzhou, China	1 January 2001 to 31 December 2005	Time-Series	Generalized Additive Models (GAM) in Poisson regression	PM ₁₀	Cardiac diseases (ICD-10: I00-I52) Cerebrovascular diseases (ICD-10: I60-I69)	Age & Sex	Percent Change (%) per IQR increase	BSk
Zhou et al. (2021)	Taiyuan, China	January 1, 2013 to October 19, 2015	Time-Series	Generalized additive model (GAM) combined with distributed lag model (DLM)	PM _{2.5} & PM ₁₀	Elderly Mortality	Sex	Percent Change (%) per 10 µg/m3 exposure increase	BSk